GP2 Early Stage Desk Research Process Mapping for Theoretical Models

Purpose of this document

This document restructures the desk research on theoretical models into a functional academic mapping that can be used directly inside GP2. The purpose is not to summarise theories, but to clarify how each model strengthens GP2 by supporting problem logic, research design legitimacy, methodological rigor, construct validity, evaluation logic, and long term credibility. The structure mirrors the logic used in the Desk Research Mapping for GP2 so that supervisors can clearly see why each theoretical element exists and how it is used within the study.

This document is intentionally pragmatic. Every model is retained for transparency of process, but each is positioned according to its function in the research. Models that strongly strengthen academic defensibility are framed as core. Models that emerged during exploration but contribute less to final rigor are framed as exploratory outputs rather than as final foundations. This approach preserves full detail while still demonstrating critical judgement.

Research process note

I conducted the theoretical desk research intentionally in three focused stages. This was a deliberate research strategy rather than an accidental sequence. The purpose of doing this in stages was to ensure academic legitimacy, to test whether the problem was real across multiple independent explorations, and to verify that the selected theoretical models and theoretical outcomes are defensible and substantiated rather than arbitrary.

The first stage focused on broad problem grounding. I explored theoretical models capable of explaining why reproducibility breaks down in applied student research environments. This stage produced system level and methodology oriented foundations such as Work System Theory, Design Science Research, FAIR, provenance logic, and Normalization Process Theory. The goal at this stage was not selection, but validation that the problem has real theoretical substance.

The second stage repeated the same core research framing but intentionally re explored the landscape using a different lens, with stronger attention to educational context, organisational behaviour, and knowledge continuity. This produced overlapping but differently framed models such as Socio Technical Systems Theory, Design Based Research, Knowledge Management theory through the SECI model, and Open Science lifecycle approaches. The methodological importance of this stage is that it revealed convergence. Different explorations continued to point toward the same underlying issues of structure, documentation, continuity, and legitimacy.

The third stage was a refinement phase. Here I explicitly evaluated the emerging theoretical landscape against the actual VCH context and the practical requirements of framework design, validation, and sustainability. This stage led to prioritisation rather than expansion. Models were assessed based on whether they directly strengthened problem diagnosis, research design legitimacy, evaluation logic, and long term continuity. The final theoretical structure is therefore the result of selection and critical filtering, not accumulation.

This staged process is methodologically meaningful. It demonstrates iteration, critical reflection, elimination of redundancy, and convergence across multiple focused explorations. It shows that the problem is real, that the theoretical models are not accidental, and that the final model set is grounded in defensible academic logic.

Overview of the three theory functions

The desk research contains three layers of function. When structured properly, they form a coherent logic rather than a list of unrelated models.

Part 1 focuses on diagnosing the problem at system level. These models explain why reproducibility breaks down in applied student research environments. They justify that the issue is structural and systemic rather than individual student failure.

Part 2 focuses on designing and legitimising the framework. These models justify the use of an artifact, the need for structured processes, and the legitimacy of iterative design and evaluation in a real world context.

Part 3 focuses on evaluating quality and ensuring long term continuity. These models define what good reuse actually means and explain how practices become embedded beyond a single cohort.

Together, the three parts support a defensible argument that the problem is real, the solution approach is academically legitimate, and the expected outcomes are defensible and sustainable.

Stage one: initial theory exploration

Framing prompt and research intention

Final Comprehensive Theory Discovery Prompt, Step 1

I am working on the research phase of my Graduation Project. The study is conducted within an applied university research lab where student teams work with real external organisations on complex challenges related to transparency, sustainability, and system resilience. Research projects are often continued across semesters by new student cohorts, but documentation practices vary widely. As a result, research decisions, data sources, and analytical steps are not always traceable or reusable, which limits verification, cumulative learning, and institutional credibility. The project focuses on improving research transparency, traceability, and reusability through the design and evaluation of a reproducible research framework that can be consistently applied by students and supervisors in such applied research environments.

Main Research Question How can an applied university research lab design and validate a reproducible research framework that makes student projects transparent, verifiable, and reusable in a consistent and institutionally reliable way

The study is situated at the intersection of applied university research, stakeholder driven innovation, and reproducibility oriented research practice. Relevant theoretical or conceptual models may originate from a wide range of disciplines and research traditions. Any model is relevant if it helps explain why reproducibility breaks down in applied research settings, supports the design of structured and traceable research processes, or explains how improvements in reproducibility can be evaluated and sustained over time.

Request Please identify five theoretical or conceptual models that are most suitable for grounding this study. For each model, explain clearly the core idea of the model, why it is relevant to reproducibility challenges in student led applied research environments, which part of the research problem it most strongly supports, and key academic sources associated with the model. The five models together should form a coherent and defensible theoretical foundation for answering the main research question.

Models identified in stage one

Model 1: Work System Theory

The core idea is that a real world research lab functions as a work system, where people perform activities using information and technology to produce outputs for specific customers and stakeholders. Reproducibility problems emerge when parts of the work system are misaligned, such as unclear roles, inconsistent information objects, unstable tooling, or missing handoffs between cohorts.

This model is strongly relevant because the problem is not only a data issue. It is a sociotechnical issue where documentation quality depends on the interaction between students, supervisors, tools, templates, time pressure, and assessment routines. Work System Theory provides a defensible way to diagnose which system elements create non traceable decisions and non reusable outputs, and why those failures repeat across cohorts.

It most strongly supports problem diagnosis, and it also supports translation of diagnosis into design requirements by forcing specification of what must change in processes, information, and roles for reproducibility to hold.

Key academic sources include Alter, 2013, in Journal of the Association for Information Systems.

Model 2: Design Science Research

The core idea is that research can generate knowledge by building and evaluating an artifact that addresses a real problem in its real context. In this study, the reproducible research framework is the artifact, and the lab is the environment where it must be built, tested, improved, and justified.

This model is relevant because the aim is not only to explain why reproducibility breaks down, but to design a usable framework that works under authentic constraints, then validate that it improves transparency, verifiability, and reuse. Design Science Research is built for applied, stakeholder linked problem solving with rigorous evaluation.

It most strongly supports framework design and evaluation, because it provides a structured logic for iterating from problem awareness, to objectives, to design, to demonstration, to evaluation, and to communication.

Key academic sources include Hevner et al., 2004, in MIS Quarterly, Peffers et al., 2007, in Journal of Management Information Systems, and Gregor and Hevner, 2013, on positioning and communicating Design Science Research contributions.

Model 3: Provenance Model

The core idea is that provenance represents structured information about what entities were used, what activities occurred, and who was involved in producing an output. Provenance creates an explicit decision trail that can be inspected later to judge reliability and trustworthiness.

This is relevant because student research often fails on traceability rather than ambition. When a later cohort cannot reconstruct how a claim was produced, which sources were used, or why a step was taken, verification and reuse collapse. A provenance model provides a precise theoretical basis for what traceability must contain, beyond vague advice such as document better.

It most strongly supports framework design and evaluation. In design, it informs what metadata and links the framework must capture. In evaluation, it allows checking whether a case file actually contains enough provenance to reconstruct and audit key outputs.

Key sources include the W3C PROV DM specification and the PROV family overview, including PROV O as an ontology expression.

Model 4: FAIR Guiding Principles

The core idea is that research outputs should be Findable, Accessible, Interoperable, and Reusable, not only for humans but also, where possible, in machine actionable ways. Importantly, FAIR applies not only to data, but also to workflows, tools, and the contextual information required for reuse.

This model is relevant because institutional reliability depends on reuse across cohorts. FAIR provides a widely recognised, externally legible standard for what reusable means in practice, so the framework can be justified beyond local preferences and aligned with broader expectations around research stewardship.

It most strongly supports evaluation and long term knowledge continuity. It provides criteria to assess whether outputs from one cohort can realistically be found, accessed, understood, and reused by the next cohort without reliance on informal oral history from supervisors.

Key sources include Wilkinson et al., 2016, in Scientific Data, plus institutional research data management guidance that operationalises FAIR in practice.

Model 5: Normalization Process Theory

The core idea is that new practices become sustainable only when people collectively make sense of them, invest in them, enact them in daily work, and then reflect on and adjust them. Normalization Process Theory explains why some improvements remain one off projects while others become routine.

This is relevant because the framework must survive cohort turnover. Even a well designed structure will fail if students see it as extra bureaucracy, or if supervisors cannot integrate it into supervision and assessment routines. Normalization Process Theory provides a theory grounded way to design adoption supports and to evaluate whether the framework is becoming embedded rather than merely piloted.

It most strongly supports long term knowledge continuity, because continuity is not only storage, it is sustained practice. Normalization Process Theory allows assessment of implementation work, including whether the framework is understandable, feasible, and actually used in the moment decisions get made.

Key sources include May and Finch, 2009.

Stage one conclusion

Together, these models cover the full logic of the main question without gaps. Work System Theory explains where reproducibility breaks down in the lived lab system and why. Design Science Research provides the rigorous build and test structure for creating a framework that fits that system. Provenance models specify what traceability must contain at the level of decision and data lineage. FAIR provides an external, defensible target for reuse quality across cohorts. Normalization Process Theory explains how the framework can become routine and durable, which is the real condition for institutional reliability over time.

Stage two: second independent theory exploration

Research intention

The second stage repeated the same core framing prompt but intentionally re explored the theoretical landscape using a different lens, with increased attention to educational context, organisational learning, and knowledge continuity. The purpose was not to replace the first stage but to test whether a new exploration would contradict or converge with the earlier findings.

Models identified in stage two

Model 1: Design Based Research

Core idea Design Based Research is an iterative methodology that aims to develop and refine practical solutions through design, implementation, and analysis in real world settings.

Relevance to applied research Student led research labs are real world learning environments. Design Based Research fits the educational context by combining theory development with practice refinement, while requiring documentation and iteration, which are essential for reproducibility and transparency.

Strongest contribution Framework design, evaluation, and long term knowledge continuity.

Key academic sources Anderson and Shattuck, 2012, and Wang and Hannafin, 2005.

Model 2: FAIR Principles

Core idea Originally developed for scientific data, the FAIR principles aim to ensure that research outputs such as data, code, and documentation are reusable by humans and machines.

Relevance to applied research Reproducibility is undermined when outputs lack standard formats or documentation. FAIR provides a universal guide to structure student research for future cohorts.

Strongest contribution Framework design, evaluation, and knowledge continuity.

Key academic sources Wilkinson et al., 2016, and Mons et al., 2017.

Model 3: Socio Technical Systems Theory

Core idea Socio Technical Systems theory proposes that successful systems must integrate both social components such as people, culture, and processes, and technical components such as tools and infrastructure.

Relevance to applied research Reproducibility fails not only due to technical gaps such as missing data, but also due to social gaps such as poor handover, loss of institutional memory, or differing supervisor expectations.

Strongest contribution Problem diagnosis and framework design.

Key academic sources Trist, 1981, and Mumford, 2006.

Model 4: Knowledge Management Theory through the SECI model

Core idea The SECI model explains how tacit and explicit knowledge are created, shared, and retained through socialization, externalization, combination, and internalization.

Relevance to applied research Applied labs often suffer from tacit knowledge loss. Students know things that are never documented. SECI supports the structuring of knowledge sharing practices and tools for institutional memory.

Strongest contribution Knowledge continuity, framework design, and problem diagnosis.

Key academic sources Nonaka and Takeuchi, 1995, and Nonaka and von Krogh, 2009.

Model 5: Open Science Framework and research lifecycle models

Core idea The Open Science Framework promotes full cycle transparency from planning and preregistration to publication and archiving, encouraging structured documentation.

Relevance to applied research Student projects typically lack structure in documenting methods, assumptions, and changes over time. OSF lifecycle approaches help embed traceability at every stage.

Strongest contribution Problem diagnosis, evaluation, and framework design.

Key academic sources Nosek et al., 2015, and Fecher and Friesike, 2014.

Stage two conclusion

The critical significance of stage two is convergence. Even though the lens changed, the models continued to point toward the same underlying issues already identified in stage one: the importance of structure, lifecycle logic, sociotechnical alignment, tacit knowledge externalisation, and institutional continuity.

Stage three: refinement and contextual alignment

Research intention

The third stage explicitly evaluated the accumulated theoretical landscape against the actual VCH context and the practical requirements of framework design, validation, and sustainability. The purpose of this stage was to move from exploration to defensible prioritisation.

Models prioritised in stage three

Model 1: Open Science Framework

The core idea is that research should be organised in structured, modular, and transparent workflows where data, materials, protocols, analysis steps, and outputs are stored in version controlled repositories. OSF promotes practices such as preregistration, linked project components, and clear provenance of files and decisions, which together enable verification and reuse.

This is highly relevant to student led applied research environments because reproducibility often fails not due to lack of effort, but due to fragmented storage, undocumented changes, and loss of rationale during handovers. In a setting like VCH, where projects involve messy external datasets and iterative experimentation, OSF style structures such as repository based projects with explicit readme files, version histories, and linked components directly counter the loss of decisions about cleaning, modelling, or interpretation across cohorts.

It most strongly supports framework design. It informs how the framework should be operationalised in practice, such as through standardised project structures, logging conventions, templates for documenting supply chain datasets, and explicit versioning of AI assisted analyses.

Key academic sources include Nosek et al., 2015, and Peng, 2011.

Model 2: FAIR Data Principles

The core idea is that research outputs should be Findable, Accessible, Interoperable, and Reusable through clear metadata, persistent identifiers, transparent access conditions, standardised formats, and explicit licensing. FAIR moves reuse from an aspiration to a set of concrete design requirements.

This is directly relevant to the VCH context, where projects can involve partner sensitive datasets such as ESG metrics, supplier information, or blockchain traces. In such cases, reuse failures often stem from missing metadata, unclear access rules, or idiosyncratic file formats rather than unwillingness to share. FAIR provides a principled way to diagnose these barriers and translate them into concrete requirements such as data dictionaries, documentation standards, and persistent references.

It most strongly supports problem diagnosis and evaluation. It provides criteria for identifying why existing outputs are not reusable and for assessing whether the framework improves findability, accessibility, interoperability, and reusability of project artifacts across cohorts.

Key sources include Wilkinson et al., 2016, and Gooch et al., 2019.

Model 3: Design Science Research Methodology

The core idea is that rigorous research can be conducted through the design, construction, and evaluation of an artifact that addresses a real world problem. Design Science Research Methodology structures this as an iterative process involving problem identification, objective definition, artifact design, demonstration in context, evaluation, and reflection.

This is highly relevant because the project does not aim only to describe the reproducibility problem but to design and validate a working framework within an authentic applied research environment. Design Science Research Methodology legitimises the framework itself as a research contribution and provides a defensible structure for iterating with stakeholders such as students, supervisors, and lab coordinators.

It most strongly supports framework design and evaluation. It provides the logic for piloting the framework in real projects, collecting evidence of improvement, refining the design, and justifying that the resulting artifact has both practical utility and research rigor.

Key sources include Peffers et al., 2007, and Hevner et al., 2004.

Model 4: Supply Chain Risk Management framing

The core idea is that complex systems such as supply chains remain resilient by systematically identifying, assessing, mitigating, and monitoring risks across actors, flows, and processes. Supply Chain Risk Management highlights the importance of visibility, traceability, and continuity of information flows to prevent systemic breakdown.

This is relevant because VCH explicitly operates in the domain of supply chain transparency and resilience. When viewed through a Supply Chain Risk Management lens, reproducibility failures are not merely academic issues but knowledge risks. Lost documentation, undocumented assumptions, and broken handovers function like disruptions in an information supply chain. This framing allows diagnosis using vocabulary that is meaningful to VCH stakeholders and aligned with their mission.

It most strongly supports problem diagnosis, because it helps map reproducibility failures to stages such as risk identification, monitoring, or closure, making the problem legible within the lab's existing conceptual landscape.

Key sources include Christopher and Peck, 2004.

Model 5: Communities of Practice Theory

The core idea is that knowledge becomes sustainable not primarily through documents, but through shared practice within a community that develops common language, tools, norms, and routines over time. Communities of practice persist even when individual members change.

This is crucial in a student turnover environment. Even a technically well designed framework will fail if each new cohort treats documentation as a bureaucratic obligation rather than part of competent research identity. VCH already shows characteristics of a community of practice through shared workshops, channels, informal knowledge exchange, and evolving toolkits. Communities of Practice theory explains how reproducibility practices can be embedded culturally rather than imposed procedurally.

It most strongly supports long term knowledge continuity. It explains how the framework can become part of how the lab operates rather than a temporary project output, and it provides a basis for designing rituals, shared templates, peer review practices, and supervision routines that sustain practice across cohorts.

Key sources include Wenger, 1998, and Wenger et al., 2002.

Stage three conclusion

The significance of stage three lies in prioritisation. Rather than accumulating more theory, this stage clarified which models genuinely strengthen the research and which are better understood as exploratory stepping stones within the research process.

Why this process strengthens academic legitimacy

Taken together, the three stages demonstrate a clear research trajectory. The theory set is not arbitrary, not generated in a single pass, and not dependent on a single lens. It is the product of iterative exploration, convergence testing, contextual evaluation, and selective refinement.

This strengthens the credibility of the theoretical grounding. It demonstrates that the problem is real, that multiple independent theoretical explorations converge on similar structural explanations, and that the final prioritised model set is defensible both academically and contextually. This directly supports the legitimacy of the research design, the defensibility of the framework, and the credibility of the conclusions that will be drawn in GP2.

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